Yixin Chen

PhD candidate, Peking University

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RESEARCH PROFILE

I work on medical image segmentation and learning across PET, CT and MRI. My clinical research focuses on cancer, fatty liver disease and multi-organ metabolism.

EDUCATION

Peking University September 2023-present
PhD in Medical Technology

King's College London 2019-2020
MSc in Artificial Intelligence, with Distinction

Xiamen University 2015-2019
BSc in Automation; second bachelor's degree in Mathematical Finance

CHALLENGE RESULTS AND RESEARCH FUNDING

autoPET V: second place in Category 1 2026
MICCAI autoPET V challenge, interactive PET/CT lesion segmentation.

CMR-MULTI: second place 2026
Multi-sequence cardiac MRI segmentation. Method paper selected for a CMRSeg challenge oral presentation.

REG²: fourth place 2026
Method paper selected for a MICCAI 2026 challenge workshop poster.

CHIMERA-agent: Innovative Agent Solution Award 2026
Team PKU Intelligence, MICCAI 2026.

NSFC doctoral research project 2026-2027, NSFC grant for doctoral students
Principal investigator of a project on extracting and analysing multi-organ metabolic information from total-body PET/CT.

NEWS COVERAGE

Doctoral student wins two awards in the MICCAI 2026 international challenges 2026-10-10
Institute of Medical Technology, Peking University Health Science Center

Yixin Chen awarded a 2025 NSFC doctoral research grant 2026-01-13
Institute of Medical Technology, Peking University Health Science Center

PROFESSIONAL EXPERIENCE

AirDoc July-September 2022
Algorithm Engineer
Developed a model to predict the gap between an implanted ICL lens and the natural lens after surgery, using eye measurements and lens size.
Worked on ultra-widefield fundus image analysis, clinical requirements and medical-device registration.

BioMind November 2020-June 2022

Medical Imaging Algorithm Engineer

Developed cardiac ultrasound models for image quality assessment, Doppler segmentation and measurement, septal defect detection and valve regurgitation analysis.

Worked with clinicians to integrate models into an ultrasound-assisted diagnosis system. Also contributed to an ovarian-tumour ultrasound system.

PUBLICATIONS AND PRESENTATIONS

2026

EJNMMI, 2026
Accepted

Whole-Body [^{18}F]FDG PET/CT Characterizes Multi-Organ Metabolic Alterations in MASLD

Yixin Chen (first author), Chuanzhi Zhou, Lijuan Guo, et al.

ICML, 2026
Spotlight

NeuronCtrl: Geometry-Aware Safe Closed-Loop Generative Control for Neuronal Microenvironment Dynamics

Haowei Xu, **Yixin Chen**, Wanyi Fu, Hongbin Han, Zhaoheng Xie
Conference OpenReview

IEEE TRPMS, 2026

Neural Network Approximation Aided Kinetic Parameter Inference of Compartment Models in Dynamic PET

Wenrui Shao, **Yixin Chen**, Ourui Fu, Tian Li, Zhi Yang, Nan Li, Hongbin Han, Xiangxi Meng, Zhaoheng Xie
Paper

CVPR Findings, 2026

PHATE-Net: Differentiable Pseudotime Learning for Trustworthy Disease Trajectories in PET

Yixin Chen, Yan Wang, Wenrui Shao, Zhaoheng Xie
Paper

2025

Nature Communications, 2025

Modality-projection universal model for comprehensive full-body medical imaging segmentation

Yixin Chen (first author), Lin Gao, Yajuan Gao, Rui Wang, Jingge Lian, Xiangxi Meng, Yanhua Duan, Leiying Chai, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie
Paper arXiv Code

Health Data Science, 2025

VP-SFDA: Visual Prompt Source-Free Domain Adaptation for Cross-Modal Medical Image

Yixin Chen, Yan Wang, Zhaoheng Xie
Paper

EJNMMI Physics, 2025

Neural network-aided unsupervised input function estimation for dual-time-window PET Patlak analysis

Wenrui Shao, Yarong Zhang, Fen Du, Fangxiao Cheng, **Yixin Chen**, Xiangxi Meng, Ying Liang, Zhaoheng Xie
Paper

IEEE NSS/MIC/RTSD, 2025

A Physiology-Supervised Neural Operator Framework for Kinetic Parameter Estimation in Dynamic PET

W. Shao, **Yixin Chen**, Z. Wang, R. Xu, X. Meng, Z. Xie
Paper

2024

IEEE TMI, 2024

PCNet: Prior Category Network for CT Universal Segmentation Model

Yixin Chen, Yajuan Gao, Lei Zhu, Wenrui Shao, Yanye Lu, Hongbin Han, Zhaoheng Xie
Paper Code

MICCAI, 2024
Poster

LUCIDA: Low-Dose Universal-Tissue CT Image Domain Adaptation for Medical Segmentation

Yixin Chen, Xiangxi Meng, Yan Wang, Shuang Zeng, Xi Liu, Zhaoheng Xie
Paper Code

IEEE TRPMS, 2024

Structure-Enhanced Unsupervised Domain Adaptation for CT Whole-Brain Segmentation

Yixin Chen (first author), Yajun Gao, Lei Zhu, Jianan Li, Yan Wang, Jiakui Hu, Hongbin Han, Yanye Lu, Zhaoheng Xie
Paper Code

IEEE NSS/MIC/RTSD, 2024

Neural Network-Asissted Dual-time-window Acquisition: Modeling Incomplete Input Functions for Patlak Analysis

W. Shao, Y. Zhang, **Yixin Chen**, X. Meng, Y. Liang, Z. Xie
Paper

2023

IEEE TMI, 2023

FVP: Fourier Visual Prompting for Source-Free Unsupervised Domain Adaptation of Medical Image Segmentation

Yan Wang, Jian Cheng, **Yixin Chen**, Shuai Shao, Lanyun Zhu, Zhenzhou Wu, Tao Liu, Haogang Zhu
Paper

IEEE EMBC, 2023

Rethinking Disentanglement in Unsupervised Domain Adaptation for Medical Image Segmentation

Yan Wang, **Yixin Chen** (first author), Yingying Zhang, Haogang Zhu
Paper

Frontiers in Cardiovascular Medicine, 2023

Echocardiography-based AI for detection and quantification of atrial septal defect

Xixiang Lin, Feifei Yang, **Yixin Chen** (first author), et al.
Paper

2022

IEEE EMBC, 2022

MSGAN: Multi-Stage Generative Adversarial Networks for Cross-Modality Domain Adaptation

Yan Wang, **Yixin Chen** (first author), Wenjun Wang, Haogang Zhu
Paper

Cancers, 2022

A deep learning model system for diagnosis and management of adnexal masses

Jianan Li, **Yixin Chen** (first author), Minyu Zhang, et al.
Paper

Frontiers in Cardiovascular Medicine, 2022

Echocardiography-based AI detection of regional wall motion abnormalities and quantification of cardiac function in myocardial infarction

Xixiang Lin, Feifei Yang, **Yixin Chen**, et al.
Paper

Preprints

arXiv, 2026

PreSight: Preoperative Outcome Prediction for Parkinson's Disease via Region-Prior Morphometry and Patient-Specific Weighting

Yand Wang, Chen Zhang, Lanyun Zhu, **Yixin Chen**, Qunbo Wang, Yutong Bai, Jurgen Germann, Yinghong Wen, Shuai Shao
arXiv

SSRN preprint, 2025

Whole-Body Inter-Organ Metabolic Atlas for PET/CT Anomaly Detection

Yixin Chen, Lin Gao, Lei Zhu, Ruoyan Xu, Fangxiao Cheng, Xinyu Liu, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie
Preprint

arXiv preprint, 2023

FP-PET: Large Model, Multiple Loss And Focused Practice

Yixin Chen, Ourui Fu, Wenrui Shao, Zhaoheng Xie
arXiv

arXiv, 2023

SL: Stable learning in source-free domain adaption for medical image segmentation

Yixin Chen, Yan Wang
arXiv

Challenge papers and conference abstracts

MICCAI REG² challenge workshop, 2026
Poster

Diagnosis-Conditioned Retrieval for Structured Pathology Workflows

Yixin Chen
Paper

MICCAI CMRSeg workshop, 2026
Selected for oral presentation

Acquisition-Structured nnU-Net Ensembles for Multi-Sequence Cardiac MRI Segmentation and Quantification

Yixin Chen, Yizhen Peng
Paper

MICCAI MVAA challenge workshop, 2026
Poster

Failure-Mode-Driven Multimodal Mitral Valve Segmentation for the MVAA 2026 Challenge

Yixin Chen
Paper

Multiorgan Metabolic Dysregulation in Alzheimer's Disease: Total-body PET/CT Reveals Systemic Network Interactions with Amyloid Burden

EANM, 2025
Oral abstract, OP-308

R. Xu, Y. Duan, L. Gao, **Yixin Chen**, H. Han, Z. Cheng, Z. Xie
Programme

PATENTS, TEACHING AND ACADEMIC SERVICE

12 invention patents

4 granted; 8 under substantive examination.

Nature Communications

Early Career Co-Reviewer

Computer-vision teaching

Recorded introductory and intermediate courses on machine learning and computer vision.

Deep Learning Theory and Practice: PyTorch Case Studies

2021

Sole author. Published by Tsinghua University Press in 2021.